

Tilak Shrivastava

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[GitHub](#) | [Leetcode](#) | [LinkedIn](#)

SUMMARY

Innovative software developer skilled in C++, Python, and React Native. Experienced in high-performance systems programming, network protocol analysis, and scalable full-stack web applications. Proven ability to collaborate effectively and solve complex problems using Git and GitHub. Want to know more? [\[Talk to my AI Bot\]](#)

PROJECTS

High-Performance Deep Packet Inspection (DPI) Engine – [GitHub Repo](#)

C++ | Multithreading | PCAP Analysis | Network Protocols (Ethernet, IP, TCP/UDP, TLS)

- Architected a multi-threaded **Deep Packet Inspection (DPI) engine in C++** to analyze network traffic and classify Layer-7 application protocols.
- Implemented **TLS Client Hello SNI (Server Name Indication) parser** to identify encrypted domain traffic (YouTube, Facebook, etc.) without decryption.
- Engineered a high-throughput pipeline featuring a **Load Balancer, Fast-Path caching, and Connection Tracker (5-tuple flow state)** for real-time packet filtering.
- Integrated dynamic **Rule Engine and PCAP Writer** to enforce real-time domain blocking, traffic policy filtering, and audit reports.

Grocery Shopping & Delivery Platform – [View](#)

React.js | Tailwind CSS | Node.js | Express.js | PostgreSQL | Inngest

- Developed a full-stack grocery shopping platform with **role-based authentication** for Admin, User, and Delivery Partner.
- Built an **Admin Dashboard** for managing products, orders, and delivery partners, along with a dedicated delivery partner dashboard.
- Implemented **real-time order tracking and live location sharing** for delivery partners.
- Integrated automated email notifications and **background jobs with Inngest** to automatically assign delivery partners to orders.

AI Website Builder – [View](#)

MongoDB | Express.js | React.js | Node.js | OpenRouter | REST APIs

- Developed a full-stack AI-powered SaaS platform that generates websites using natural-language prompts via **OpenRouter AI APIs**.
- Implemented user authentication with signup/login and built a **manual code editor** allowing users to customize AI-generated code.
- Implemented **website export functionality** and designed a scalable **MERN architecture** with production-oriented project structure.

SKILLS

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|---|---|
| • Technical skills & languages: C++, JavaScript, Python, SQL | • Networking & Systems: TCP/IP, TLS SNI, PCAP Parsing, Multithreading |
| • Frontend development: HTML5, CSS3, React.js, Tailwind CSS | • CS fundamentals: Data structures, Algorithms, OOP, DBMS, OS, Computer networks |
| • Backend development: Node.js, Express.js, REST APIs | • Other skills: MERN stack, JWT authentication, Responsive web design |
| • Database management: MongoDB, MySQL, SQLite, PostgreSQL | • Tools and platforms: Git, GitHub, VS Code, Postman, Vite, Wireshark |

EDUCATION

Master of Computer Applications

Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore | CGPA: 7.67